Customer Shopping Behavior Analysis

End-to-End Data Analytics Project

Tools Used

- Python (pandas)
- SQL (PostgreSQL)
- Power BI

Data Analytics Project Pipeline

Key Insights

What the data revealed (3,900 purchases)

Highlights

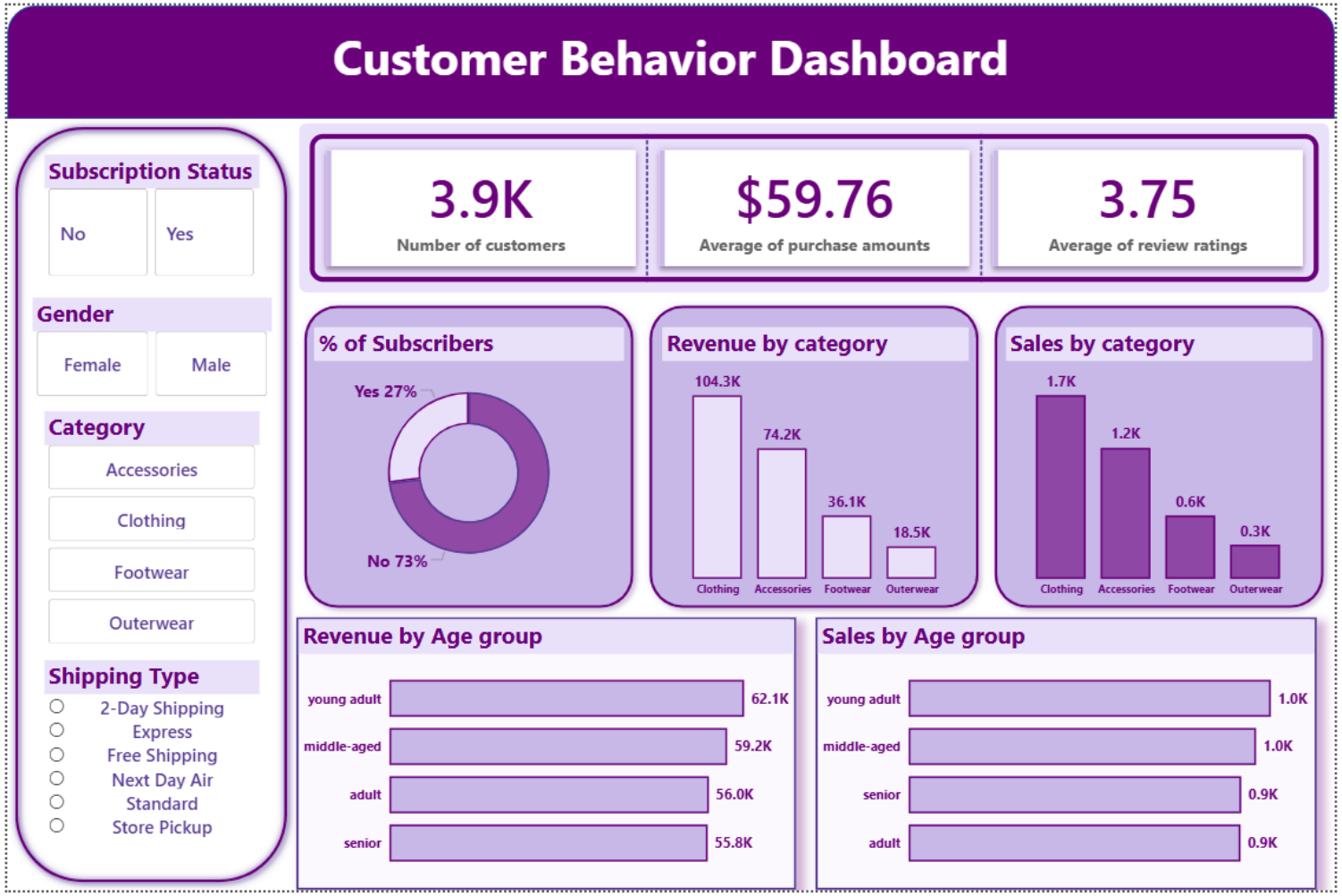
- Clothing category generates the highest revenue and sales.
- Subscription adoption is low (27%) → strong growth opportunity.
- Young Adult & Middle-aged groups contribute the most revenue.
- Shipping & discounts influence spending behavior.

KPI Summary

3.9K Total Purchases	**$59.76** Avg Purchase Amount
3.75 Avg Review Rating	**27%** Subscribers

Power BI Dashboard

Interactive dashboard built for stakeholders



Business Recommendations

Data-driven actions for growth

Recommended Actions

- Boost subscriptions with exclusive rewards & benefits.
- Build loyalty programs to convert Returning → Loyal customers.
- Review discount-dependent products to protect profit margins.
- Focus promotions on top categories and high-value segments.

Before vs After Data Cleaning

BEFORE (Raw Data)

- Missing values (Review Rating)
- Inconsistent column names
- Mixed data types
- Redundant columns present
- Not ready for BI dashboards

AFTER (Clean Data)

- Missing values imputed (median by category)
- Column names standardized (snake_case)
- Feature engineering added (age_group etc.)
- Redundant column removed (promo_code_used)
- Prepared for SQL + Power BI analysis